

CITS5508 Practice Exam — Chapter 6

Decision Trees

- This practice paper has **5 questions** for **60 marks**.
 - Allow approximately **two minutes per mark** (≈ 2 hours total).
 - Only UWA-approved calculators permitted.
 - Answers are in the separate file *Practice Exam - Chapter 6 (Answer Key).md*.
-

Question 1. (12 marks)

A decision-tree node applies to **50** training instances with class counts: **0** of class A, **40** of class B, **10** of class C.

- (a) Compute the node's **Gini impurity**. Show your working. (3 marks)
 - (b) Compute the node's **entropy** (use $\log_2$). Show your working. (4 marks)
 - (c) What would the Gini impurity be if all 50 instances belonged to a single class? What do we call such a node? (2 marks)
 - (d) State one practical difference between using **Gini impurity** and **entropy** as the split criterion, and say which is the scikit-learn default and why. (3 marks)
-

Question 2. (12 marks)

The **CART** algorithm is considering a split of a node containing **100** instances. The candidate split produces: - **Left** child: 40 instances — 30 of class A, 10 of class B. - **Right** child: 60 instances — 10 of class A, 50 of class B.

- (a) Compute the Gini impurity of the **left** child and of the **right** child. (4 marks)
 - (b) Compute the **CART cost** for this split (the size-weighted average impurity of the children). (3 marks)
 - (c) Explain why CART is described as a **greedy** algorithm, and why finding the optimal tree is generally intractable. (3 marks)
 - (d) State the time complexity of **prediction** and of **training** for a decision tree (in terms of m instances and n features). (2 marks)
-

Question 3. (12 marks)

- (a) Decision trees are called **nonparametric** models. Explain what this means and why it makes them prone to **overfitting**. (4 marks)
 - (b) List **four** hyperparameters you can use to **regularise** a `DecisionTreeClassifier`, and for each state whether you increase or decrease it to reduce overfitting. (4 marks)
 - (c) Briefly describe **pruning** as an alternative regularisation approach, including the role of a statistical test such as the χ^2 test. (4 marks)
-

Question 4. (12 marks)

- (a) Decision trees are described as **white-box** models, whereas random forests and neural networks are **black-box**. Explain the distinction and why it matters. (4 marks)
- (b) Explain how a decision tree **estimates class probabilities** for a new instance. Why does it give the **same** probability estimate for every instance that falls in the same leaf?

(4 marks)

(c) Decision trees require **very little data preparation**. Name one preprocessing step that is **not** required for decision trees but **is** required for SVMs / k-NN, and explain why.

(4 marks)

Question 5. (12 marks)

(a) Explain the **sensitivity to axis orientation** limitation of decision trees, and describe one technique that mitigates it. (4 marks)

(b) Decision trees are said to have **high variance**. Explain what this means and how an **ensemble** (random forest) addresses it. (4 marks)

(c) Write a short scikit-learn snippet that trains a `DecisionTreeClassifier` regularised with `max_depth=3` and `min_samples_leaf=10`, and prints the trained tree's feature importances. (4 marks)